

Automated Assessment Tools for grading of programming Assignments: A review

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Abstract— Programming assignments play a key role in computer science educational curriculum. Universities use substantial techniques to enhance the practical programming skills of students. This require assignments and assessment to be planned systematically. Assessment of programming assignments provide a mechanism to impact students' learning outcomes as well as approaches for instructors who evaluate the programming assignments. Manual assessment is time-consuming and therefore Automated Assessment Tools (AATs) are on demand to evaluate programming language assignments. This study gives a thorough evaluation of several Automated Assessment Tools that have been developed so far to evaluate programming assignments for the implementation of specific algorithms. We have discussed the features and the approaches used by various tools developed during different years. The goal is to conduct a comparative study among different Assessment Tools and presents the reason to develop a new Automated Assessment model.

Keywords—Computer programming; Evaluation of Programming Assignments; Assessment; Grading; Automated Assessment Tools;

I. INTRODUCTION

The programming courses emphasize learning different programming constructs and syntax as well as building different skills like problem-solving ability, abstract thinking capabilities, logic building, and gaining confidence in mathematical and scientific methods. A variety of assessments are available to assess the programming knowledge of the students such as programming assignments to check student's logic, their ability to explain what they are doing, and they code with correct syntax and output [1].

The objective of this paper is to elaborately present different Automated Assessment Tools (AAT) which are developed in the past and currently available.

This survey will benefit instructors who grades large classes. Automatic marking of programming assignments might save the instructor and students a lot of time, allowing them to get grades faster and become more self-aware after receiving immediate feedback and practice to advance their programming abilities.

A. Automated Assessment system: A solution

In the manual assessment technique, programming assignments are graded manually by the instructor. So, feedback and grading are delayed. Manual evaluation is labor-intensive and time-consuming when no. of students are more. Only a few programming tasks may be given out due to the time commitment. The manual verification of each evaluation will be slower and of lesser quality [2]. Hence, automated tools can be used by educational institutions for the assessment of programming knowledge and skills. To overcome such concerns, Computer-Aided Assessment (CAA) or Automated Assessment Tools (AAT) were developed [3].

The AATs analyzes the code for syntactic errors, verify its proper implementation, coding style, the programming ability, and the problem-solving skills and provides instant feedback to both the instructor and the student [4]. Such automated tools do not limit the number of assignments provided by the instructor and very helpful in Massive open online courses (MOOC). Typically, Assessment is testing the program structurally and functionally [5]. From both a pedagogical and a technical standpoint, this article describes the primary aspects that the tools provide, as well as the various techniques they employ.

B. Approaches Implemented in Automated Programming Assessment

Programming assessment can be on Multiple Choice Questions (MCQs) or subjective with code testing. Continuous formative assessment is needed to enhance learning and to assist students to evaluate their own learning. Static and dynamic are two approaches used in programming assessment (Ala-Mutka, 2005) [4]. Most of the AATs (Automated Assessment Tools) available implement either static or dynamic analysis approach or both approaches to complement each other (Rohaida, 2006).

- **Static Analysis:** Static analysis is a technique test a code without having to run it. Programming style evaluation [6] can take several forms, including the testing of different metrics in software, structural and similarity analysis, detecting plagiarism and diagrams. Infinite loops, logical mistakes, and unused statements are all discovered using this method.

- Dynamic analysis: Dynamic analysis entails the execution of code. The program's functioning is assessed to see whether it meets a minimum standard. The accuracy and efficiency of the code are evaluated using dynamic analysis [7].

II. AUTOMATED ASSESSMENT TOOLS USED IN THE ASSESSEMENT OF PROGRAMMING LANGUAGE

From a pedagogical and technological standpoint, this article addresses the primary aspects that the tools enable, as well as the various technique they employ. In this study, the benefits and drawback of various AAT are discussed.

- 2TSW [8] a web based gamified AAT developed in 2019 for the automated correction of C programming task. Students are given a set of programming homework by 2TSW. The system uses a set of unit tests developed by the trainer to evaluate the student's suggested solutions for each assignment in real time. The evaluation procedure generates the number of exams passed as well as details feedback on the test that were failed and their scores.
- PAAA [9], Programming Assignments Automatic Assessing (PAAA), is a grammar analysis tool that uses ANTLR, a parser generator. A program that aids in the design of parsers. A parser converts a piece of code into a logical structure, as in an Abstract Syntax Tree (AST).
- PCQL [10], Parameterized Code Query Language, which not only dynamically tests but also statically evaluates a student's program based on certain custom rules presented by the instructor, thereby automating the process of evaluation, and removing any possible bias and error.
- DANTE: [11] This is an Automated assessment Tool designed for a university developed by P. Duch, T. Jowrki and Dante in 2018. It automatically grades the student's programming assignments using acceptance testing and unit testing. It is designed as both a standalone and web application. The System evaluates the correctness of student's submission, written in C language. Students can send both single and multiple solutions.
- The estimation of student learning outcomes is evaluated by a model called IRT-BKT models [12]. The Item Response Theory estimates the student's initial programming knowledge status. The Bayesian Knowledge Tracing (BKT) checks the probability of knowing a skill before training, the probability of acquiring the skill from unknowning, the probabilities of guessing without knowing the skill and failing despite knowing the skill.
- NCV (Ninja Code Village) [13] is a full visual programming language learning help environment. It also assesses computational thinking principles like conditional statements, loops, data, and parallelism automatically to help students improve their programming abilities.
- INGINIUS [14] is a tool, which can virtually assess many kinds of programming languages.
- Code checker is a method for testing any preset problem in an automated manner. This is developed by P.R.Choudhury, N.Wats, R.Jaiswal, R H Goudar in 2014. It displays the minor errors, the correct answer, the partially correct answer, and wrong answers in the program submitted by a student.
- JavaBrat is an auto-grading solution that benefits Java and Scala coders [15]. JavaBrat is integrated into several online learning platforms, such as Moodle. Javabrat, displays the programming issue, accepts user solutions, and provides the user with a grade and feedback.
- The Automatic Grading System (AGS) [30], AGS may analyze, grade, and manage the programming exercises of the students. They propose a Flexible Dynamic Analyzer that uses the Output Semantic-Similarity Language to do semantic similarity analysis (OSSL).
- JAssess is a solution that allows you to organize and assess students' Java programming exercises that they submit using MoodleTM. It displays the recommended mark together with the outcome for each successful compilation, and the defect along with the suggested mark for each failed compilation.
- JUnit Generation (JUG) [17] offer learners with rapid, semi-automated feedback. It generates unit tests and time complexity tests using a Java-like script, then executes those tests to provide results. Improved feedback for students and reduced preparation and grading time for instructors and grading helpers are the aims of JUG.
- A Programming Exercise, Petcha [18] an "automated Teaching Assistant," assisting students in learning to code more quickly. It aids educators in the creation of programming activities that are evaluated automatically. This gives information on the many types of errors that students make in their submissions.
- SQL-based paperless examination system [19] evaluates the SQL queries. It employs a mechanism for checking the validity of DML statements with return set and empty return set, ensuring that SQL queries are correctly verified both syntactically and semantically.
- The Web-CAT, an automated grading system that permits students to submit several versions of an assignment and receive instant feedback. Correctness, coding style, and code coverage by student-written tasks are measured [37].
- APOGEE [20] may offer step-by-step playback coaching, giving useful feedback to help students make thoughtful and iterative changes in their learning.
- JThreadSpy [21], a tool gives a view of a multithreaded Java program's execution trace with a customizable degree of granularity. This tool directly illustrates the connections between distinct threads,

which might only be deduced implicitly via the use of a debugger or static code analysis.

- BOSS [22] help with online programming assignment submission and subsequent processing. Students can use this tool to get comments on their submitted programs, and teachers can use it to assess student submissions. Static analysis and preliminary validation are used.
- Marmoset [23] is a framework that allows you to submit and test your work automatically. Every time a student saves a file, this system takes a picture of their work. This information is important to researchers who may be looking at how learners learn programming languages.
- GAME [Blumenstein, M., Green, S., Nguyen, A. and Muthukkumarasamy, V.,2004] automatically analyze programming assignments written in a range of languages based on the source code's "structure" and the validity of the program's output. Now, the system can label programs written in Java, C++, and C. GAME has been put to the test in a variety of programming activities and projects for students. The collected data were examined and compared to a human marker, yielding promising results.
- nExaminer [Zheng Cheng] is a framework for the semi-automated assessment of computer programming assignments in Moodle. This challenge can be mitigated by providing a tool that assists both the student (via feedback on progress) and the instructor (through assistance with assignment revision).

There are so many other tools also available which uses different approaches to assess the programming exercises. Automated assessment tools are commonly used to grade students' programming assignments. Many AATs (Automated Assessment Tools) use black box testing to evaluate student programs, and then use the output comparison approach to establish the program's correctness by comparing the program output strings to the expected output pre-defined by the instructor or developed from a previous model. Some of the AATs have the capability to detect plagiarism and generate instructive feedback. Table.1 describes an overview of some of the popular Automated Assessment tools.

TABLE I. COMPARISON OF VARIOUS AUTOMATED ASSESSMENT TOOLS

Sl. No	Name	Year	Language	Functionality
1	2TSW	2019	C,C++	Testing Based Approach, Static Analysis,Gamified platformFeedback generation
2	PAAA	2018	All prog. Language	grammar analyze tool ANTLR, Dynamic analysis,multiple submission
3	PCQL	2018	All prog. Language	Custom rules, Static and dynamic analysis, Higher coding standard
4	DANET	2018	C	Functional testing, Static and dynamic analysis, code completeness
5	IRT-BKT	2017	C	Static Analysis, Plagiarism detection, Work with different feedback le

				vels
6	Petcha	2012	Eclipse	Test case based, Dynamic Analysis,
7	Webcat	2008	Java,C++	Pattern Matchin, Functional Testing, semi automatic

III. EFFECTIVENESS OF AUTOMATED ASSESSMENT TOOLS

There are several parameters for students to be discussed like, Does the AAT grade the assignment correctly? Does the AAT lay a positive impact on student learning? Does the AAT improve their performance? Does the AAT generate immediate feedback? The effect of the automated assessment tool has a positive effect on the student. After automatic testing, the correctness and quality have been increased effectively for programs. As their assignments are examined, the students are satisfied.

Instructors also get benefitted from the automated assessment tools. As manual assessment takes a lot of time and effort with a larger volume of students, Automated assessment tools check assignments faster and save time so that teachers can do other useful works related to student development.

IV. LIMITATION OF AUTOMATED ASSESSMENT TOOLS DESIGNED EARLIER

Some of the Automated Assessment tools discussed in this paper uses a static analysis technique for parameters like software metrics, quality, and functionality. Whereas Some of Automated Assessment do not provide any feedback to the student after the grading is done. Some tools only check syntax error at compiled time. The systems aren't meant to provide real-time automated evaluation, which might lead to a higher degree of personal involvement again

V. FUTURE WORK

An automatic assessment task learns how an instructor evaluates a task. The instructor assesses it based on a sample set of student replies. The approach builds a computer model based on inferred rules regarding the instructor's grading actions. The model is then used to grade other students' programming assignments. A new automated system can be designed to assess the programming assignments of students in real time and deployment of automatic feedback systems to enhance the student's programming skills and develop algorithms for outcome assessment of the students. The automated system should be language independent to conduct both static and dynamic analysis. The system should be analyzing the previous work of the student and predict the programming ability of the student so that the instructor can decide how much attention should be given to the student for better understandability.

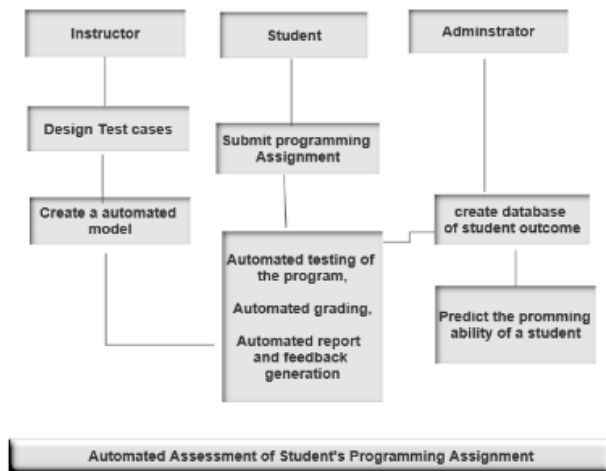


Figure 1. Architecture of future AAT

VI. CONCLUSION

The goal of this review is to look at an automated grading system that can analyze and score the source code in a student's programming assignment, as well as the tools that can help with automatic grading of student programming assignments. Automated assessment tools aid instructors in conducting a constructive evaluation process, as well as learners in learning from their mistakes and receiving fast feedback and a score. However, more research can be done in the automated assessment tools which combinedly assess the static and dynamic analysis of the code, checks plagiarism, enhance the student's programming skills by continuous feedback, developing algorithm for outcome assessment of the students, developing an algorithm to Predict the programming ability of student, developing algorithm to Predict the failure and success rate of the class.

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